

SAP CPI Project Documentation

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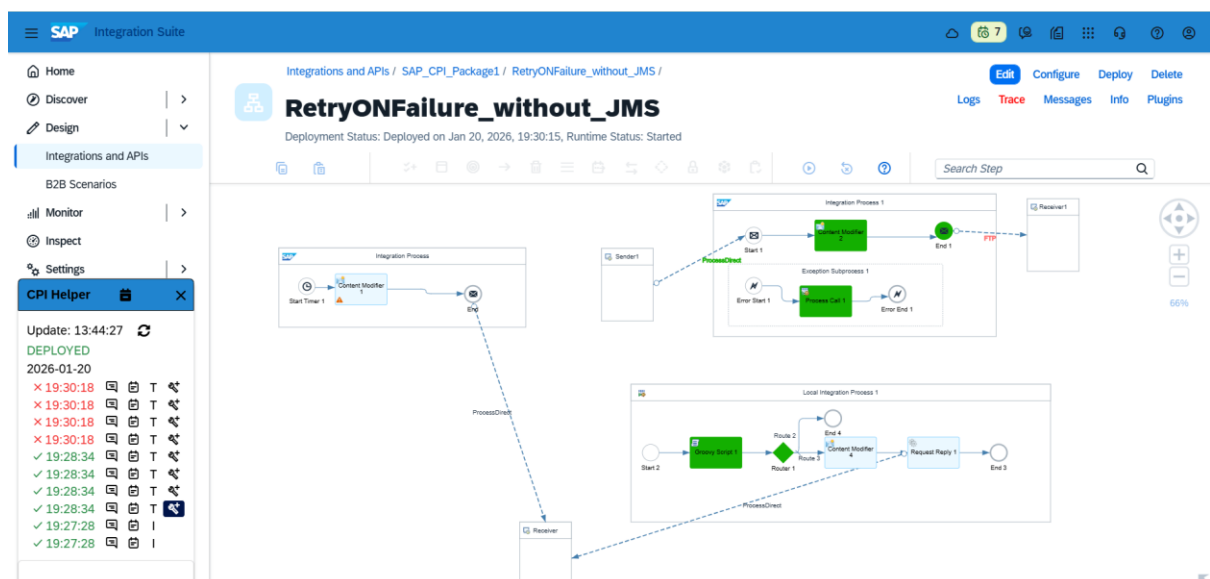
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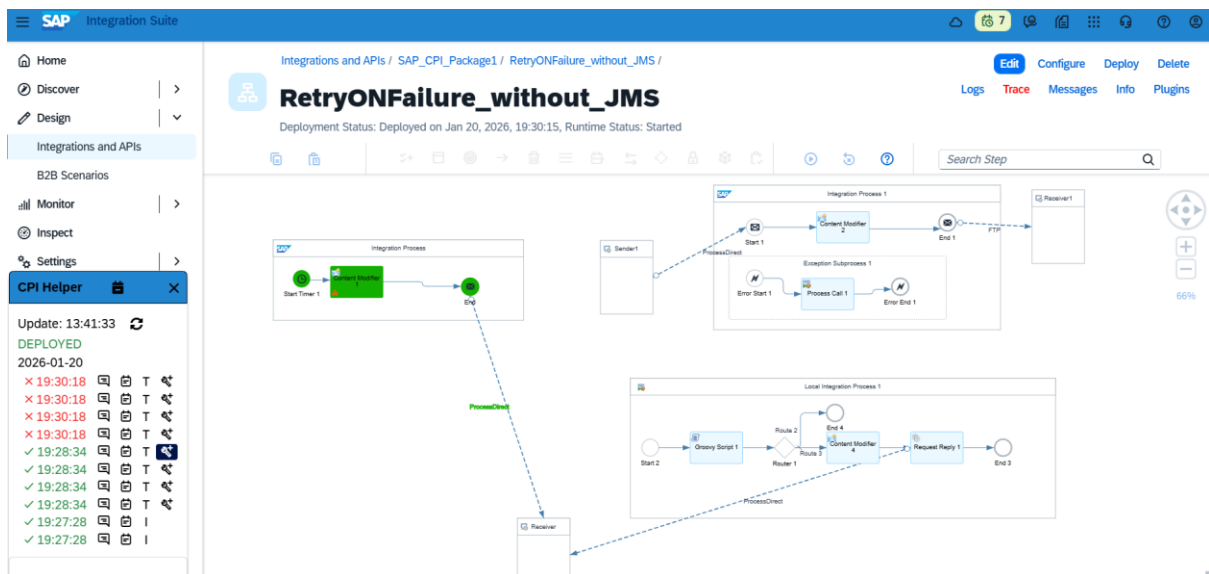
Scenario: Custom_Retry_Mechanism_without_JMS

1.Scenario Overview

This project implements a custom retry mechanism in SAP Cloud Platform Integration (CPI) to handle transient connectivity failures without using the JMS adapter. Due to limited availability of JMS queues in the CPI free/trial environment, an alternative design was implemented using Process Direct adapters, Exception Subprocesses, and Local Integration Processes. The solution ensures controlled retries with configurable retry count and time intervals, improving reliability while adhering to platform constraints.

2.Architechture Diagram





3.Step-by-Step Implementation

- Implemented a main integration flow to initiate data transmission to the receiver system.
- Designed a secondary integration flow connected via Process Direct adapter to handle outbound communication with the target system.
- Configured an Exception Subprocess to catch connectivity and runtime errors during receiver communication.
- Implemented a Local Integration Process to manage retry logic and reprocessing of failed messages.
- Used Process Direct adapters to re-invoke the main integration flow from the retry process.
- Implemented controlled retry logic using exchange properties to track and limit the number of retry attempts as per client requirements.
- Implemented time-based delay between retries using Groovy scripting to allow the receiver system to recover before reattempting message delivery.
- Ensured proper message context handling by preserving payload and relevant properties across retry attempts.
- Performed end-to-end testing to validate retry behaviour for various failure scenarios, including connectivity outages and timeout errors.

4.Adapter Configuration:

Sender: We used a Timer

Receiver: FTP Adapter

FTP

General

Target

Processing

FILE ACCESS PARAMETERS

Directory:

Retries

File Name:

/files

Append Timestamp:

☒

CONNECTION PARAMETERS

Address:

ftp.drivehq.com

Proxy Type:

Internet

Encryption:

Explicit FTPS

Credential Name:

FTP_connection

Timeout (in ms):

10000

Maximum Reconnect Attempts:

3

5.Integration Components:

Content Modifier:1

Content Modifier

General

Message Header

Exchange Property

Message Body

Headers:

Action	Name	Source Type	Source Value	Data Type	Default Value
Create	count	Constant	0		

Content Modifier:2

Content Modifier

General

Message Header

Exchange Property

Message Body

Properties:

Action	Name	Source Type	Source Value	Data Type	Default Value
Create	Payload	Expression	\$(in.body)		

Router:

Router

General

Processing

ERROR HANDLING

Throw Exception: ☐

ROUTING CONDITION

Order	Route Name	Condition Expression	Default Route
1	Route 3	\${property.retryonFailure} = 'Yes'	☐
2	Route 2		☒

Content Modifier:4

Content Modifier

General

Message Header

Exchange Property

Message Body

Type: Expression

Body: \${property.payload}

Preview

6.Groovy Script:

RetryONFailure_without_JMS / script1.groovy /



script1.groovy

```
1  /* Refer the link below to learn more about the use cases of script.
2  https://help.sap.com/viewer/368c481cd6954bdfa5d0435479fd4eaf/Cloud/en-US/148851bf8192412cba1f?
3
4  If you want to know more about the SCRIPT APIs, refer the link below
5  https://help.sap.com/doc/a56f52e1a58e4e2bac7f7adbf45b2e26/Cloud/en-US/index.html */
6  import com.sap.gateway.ip.core.customdev.util.Message;
7
8  def Message processData(Message message) {
9
10     def body = message.getBody();
11     def map = message.getHeaders();
12     def countvalue = map.get("count");
13
14     int i =countvalue as Integer;
15
16     String retryonFailure;
17     if (i<2)
18     {
19         retryonFailure = "Yes";
20         i=i+1;
21     }
22     else
23         retryonFailure = "No";
24
25     message.setHeader("count", i);
26     message.setProperty("retryonFailure", retryonFailure);
27     return message;
28 }
29
```

7. Input XML

```
<?xml version="1.0" encoding="UTF-8"?>

  <EmployeeID>EMP-1005</EmployeeID>

  <FirstName>Jane</FirstName>

  <LastName>Smith</LastName>

  <JobTitle>Senior Software Engineer</JobTitle>

  <Department>IT</Department>

  <HireDate>2015-06-15</HireDate>

  <Salary>85000.00</Salary>

  <Currency>USD</Currency>

  <Email>jane.smith@company.com</Email>

  <Phone>+1-555-987-6543</Phone>

</Employee>
```